

On Quantum Vacuum Energy, Cosmological Constant and Missing Mass

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Abstract

This paper does not propose a model. It does not present calculations. It does not claim to solve any problem. It would be presumptuous to suggest, in a short paper, answers to questions that the best minds in physics have worked on for a century. Instead, we ask a sequence of questions—each grounded in established physics—and observe that the answers lead in an unexpected direction. The questions concern three entities: quantum vacuum energy (whose existence is experimentally confirmed), the cosmological constant (whose value is observationally measured), and the missing mass in galaxies (whose presence is gravitationally inferred). We ask why the first was identified with the second, whether this identification is necessary, and what follows if it is not. We find that separating them leads, through a chain of reasoning that invokes only $E = mc^2$ and general relativity, to the question of whether the missing mass might be the gravitating vacuum itself. We do not answer this question. We ask why it is not being asked.

Keywords: vacuum energy, cosmological constant, missing mass, dark matter, dark energy, general relativity

1 A Preliminary Remark

The authors of this paper are not under the illusion that they can resolve, in a few pages, problems that have occupied the finest physicists of three generations. We are not proposing a theory. We are not constructing a model. We are not fitting data.

We are asking questions.

The questions are simple. The physics they invoke is standard. The logical chain they form is short. And we find it surprising that this particular chain of questions does not appear to have been followed to its conclusion in the literature—not because the conclusion is necessarily correct, but because the questions themselves seem unavoidable.

We ask the reader’s patience, and nothing more.

2 What Do We Know?

Let us begin with what is established.

2.1 Vacuum Energy Exists

The quantum vacuum has energy. This is confirmed by:

- The Casimir effect: a measurable force between conducting plates arising from the modification of vacuum modes (Casimir 1948; Lamoreaux 1997; Bressi et al. 2002).
- The Lamb shift: a splitting of hydrogen energy levels caused by virtual particle fluctuations (Lamb & Retherford 1947).
- Thermonuclear cross-sections: fusion rates that require virtual particle exchange for quantitative agreement with observation (Bahcall & Pinsonneault 2004).

These are not theoretical predictions awaiting confirmation. They are measurements.

2.2 Energy Is Mass

$E = mc^2$ (Einstein 1905). If the vacuum has energy, it has an equivalent mass. This is special relativity.

2.3 Mass Gravitates

The stress-energy tensor $T_{\mu\nu}$, which includes all forms of energy, is the source of gravity in general relativity (Einstein 1915). There is no exemption clause.

2.4 Vacuum Energy Depends on Geometry

The Casimir effect demonstrates that the vacuum energy density depends on the geometry of the space it occupies. Different plate separations yield different energies. More

generally, quantum field theory in curved spacetime shows that the vacuum state depends on the background geometry (Birrell & Davies 1982; Parker & Toms 2009). This is the theoretical foundation of Hawking radiation (Hawking 1974, 1975): the vacuum near a black hole horizon is different from the vacuum far away.

3 What Was Assumed?

In 1917, Einstein introduced the cosmological constant Λ as a geometric term in the gravitational field equations. It was a property of spacetime—constant, universal, geometric. He did not connect it to quantum physics.

In 1967, Zel’dovich observed that the quantum vacuum stress-energy tensor has the same mathematical form as a cosmological constant: both are proportional to $g_{\mu\nu}$. He proposed that they are the same quantity: $\Lambda = 8\pi G \rho_{\text{vac}}$.

This identification was motivated by mathematical elegance. It was never derived from a deeper principle. And it led to the cosmological constant problem: quantum field theory estimates $\rho_{\text{vac}} \sim 10^{76} \text{ GeV}^4$, while observations give $\rho_{\Lambda} \sim 10^{-47} \text{ GeV}^4$ —a discrepancy of 120 orders of magnitude (Weinberg 1989).

Our first question:

Is this identification necessary? What compels us to equate a geometric constant introduced in 1917 with a quantum field quantity proposed fifty years later?

We note that the identification has not been confirmed by any experiment. It has not been derived from any fundamental theory. Its sole motivation is mathematical similarity. And its sole consequence has been a problem that remains unsolved after sixty years of effort.

4 What Do We Gain from Dark Energy and Dark Matter?

Modern cosmology introduces two entities beyond the Standard Model and general relativity:

Dark energy is invoked to explain the accelerating expansion of the universe. It constitutes approximately 68% of the energy budget. In the standard model, it is identified with the cosmological constant Λ —which is then identified with vacuum energy, producing the 10^{120} problem.

Dark matter is invoked to explain the missing mass in galaxies (Zwicky 1933; Rubin et al. 1980), gravitational lensing (Clowe et al. 2006), the CMB acoustic peaks (Planck Collaboration 2020), and large-scale structure. It constitutes approximately 27% of the energy budget. It has not been directly detected despite four decades of dedicated searches (LUX, XENON1T, PandaX, LZ, ADMX, LHC).

Together, these two entities account for 95% of the universe. Neither has been directly observed. Both require physics beyond what is established.

Our second question:

*Do these entities make our understanding simpler, or more complicated?
Do they resolve puzzles, or create new ones?*

Dark energy, in the standard account, *is* the cosmological constant, which *is* vacuum energy—producing a 10^{120} discrepancy that remains unresolved. Dark matter requires a particle that has eluded detection for forty years and that has no place in the Standard Model.

We are told that 95% of the universe consists of things we cannot see, cannot detect, and cannot explain from known physics. This is not a criticism of the physicists who have worked on these problems—they are among the greatest scientists in history. It is an observation that the current framework, for all its successes, purchases them at a high price.

Our third question:

Is this price necessary? Or does it follow from an assumption—the Zel’dovich identification—that we are free to revisit?

5 What Happens If We Separate Them?

Suppose we take the separation seriously. Λ is geometry. Vacuum energy is field physics. They are not the same.

Immediately, several things change.

5.1 The Cosmological Constant Problem Is Reframed

If $\Lambda \neq 8\pi G \rho_{\text{vac}}$, there is no reason for them to have the same value. The 10^{120} discrepancy is a comparison of two unrelated quantities. The “problem” was created by the identification and takes a different form without it.

Of course, one might ask: why doesn’t the enormous vacuum energy gravitate cosmologically? This is a fair question, and we do not answer it. We note that it is a different question from the cosmological constant problem as traditionally stated. The CC problem asks: why do Λ and ρ_{vac} disagree by 10^{120} ? If they are unrelated quantities, this question loses its force. But the question of why a large ρ_{vac} does not dominate cosmological expansion remains—it is reframed, not eliminated. It may have a separate answer (see, e.g., the vacuum sequestering programme: Kaloper & Padilla 2014), or it may indicate that the naïve QFT estimate of ρ_{vac} is simply wrong, as many physicists suspect.

5.2 Dark Energy Becomes Unnecessary

If Λ is simply what Einstein said it was—a geometric constant, a property of spacetime—then the term “dark energy” becomes a label for geometry rather than a name for a new substance. This is more of a conceptual reframing than an empirical gain: the

physics of accelerating expansion is unchanged. But it removes the suggestion that an unknown form of energy pervades the universe, replacing it with the simpler statement that the geometry of spacetime includes a constant term. Whether this constitutes explanatory progress is a matter of perspective.

5.3 An Unexpected Question About Vacuum Energy

With Λ accounted for (geometry), we are left with vacuum energy as a separate, local, experimentally confirmed quantity. And we face the question we raised in Section 2: it has energy, therefore mass, therefore gravity.

If it is not Λ , and if it gravitates, then *where does its gravity show up?*

6 Where Does Vacuum Gravity Show Up?

Consider two environments.

6.1 Inside a Galaxy

A galaxy is a gravitationally bound system. Its interior does not expand with the Hubble flow (this is standard physics; see Carrera & Giulini 2010). The stars are separated by light-years of space filled with quantum vacuum.

If vacuum energy gravitates, this vacuum contributes mass to the galaxy. A star in orbit responds to the total enclosed mass—stars, gas, and vacuum. A lensed photon is deflected by the total mass. Neither distinguishes between sources.

6.2 In a Void

A cosmological void also contains vacuum. But here the vacuum is uniform—there is no density peak, no center of attraction, no preferred direction for collapse. And the cosmic expansion (driven by Λ) stretches the void, further opposing any tendency toward self-gravitation.

Our fourth question:

Is it possible that vacuum gravity is observationally significant inside bound structures (where expansion is absent) and observationally negligible in voids (where expansion compensates and no structure can seed)?

We do not know. We are asking.

A related open question concerns galaxy cluster mergers. In the Bullet Cluster (Clowe et al. 2006), the gravitating mass is spatially offset from the baryonic gas. If vacuum energy density depends on local spacetime curvature, it would need to track the evolving gravitational potential during a merger in a way that reproduces the observed lensing morphology. Whether it does is an open quantitative question.

7 A Coincidence

If the answer to the question in Section 6 is yes—if vacuum energy gravitates measurably inside galaxies but not in voids—then the gravitating vacuum would have the following properties:

- It gravitates.
- It is invisible.
- It does not interact electromagnetically.
- It does not scatter off itself (fields superpose; they do not collide).
- It fills all of space.
- It is in its ground state (cold, stable).

These are the properties of the missing mass.

We present this as a coincidence, not a proof. But it is a coincidence that involves every qualitative property required of the missing mass, matched by a quantity whose existence is experimentally confirmed.

We must state clearly the central vulnerability of this line of reasoning: the qualitative property match is necessary but not sufficient. The entire question hinges on a number that is currently unknown—the gravitating vacuum energy density. If this density is of the order required by galactic dynamics ($\sim 0.3 \text{ GeV}/\text{cm}^3$ in a typical halo), the coincidence of properties becomes physically significant. If it is many orders of magnitude smaller, the properties match but the quantity is irrelevant. We do not know which is the case. This is the single largest uncertainty underlying every question this paper asks.

Our fifth question:

Why has this coincidence not been investigated more thoroughly? Is there a physical reason to exclude gravitating vacuum from the list of dark matter candidates, or only the historical reason that vacuum energy was pre-assigned to a different role (the cosmological constant) by the Zel'dovich identification?

We note that several authors have explored related ideas: Albareti & Maroto (2014) showed that vacuum energy of massive fields can scale as nonrelativistic matter and reproduce cold dark matter clustering behavior—the most quantitatively developed treatment to date; Nieuwenhuizen (2024) proposed vacuum condensation on mass concentrations; Marongwe (2021) derived dark matter-like effects from electromagnetic vacuum fluctuations; Hajdukovic (2012) proposed gravitational polarization of the vacuum. None of these has entered the mainstream discussion.

The present paper differs from these in scope and intent: it does not introduce a mechanism or a model. It traces the question to what we believe is its root—the Zel'dovich identification of 1967—and asks whether removing this identification makes the question of gravitating vacuum unavoidable. We ask why this particular logical path has not been followed in the mainstream literature.

8 What We Are Not Saying

We wish to be precise about the limits of this paper.

We are not claiming that dark matter is vacuum energy. We are asking whether there is a reason it cannot be.

We are not solving the cosmological constant problem. We are observing that it may rest on an unproven assumption.

We are not deriving rotation curves, lensing profiles, or CMB spectra. We are noting that the qualitative properties match and asking whether a quantitative investigation is warranted.

We are not criticizing the physicists who built Λ CDM. Their achievement is extraordinary. We are asking whether one of the foundational identifications—made in 1967, never proven—might be worth revisiting in light of the persistent failure to detect dark matter particles and the persistent failure to resolve the 10^{120} discrepancy.

We are not proposing new physics. We are asking whether the existing physics— $E = mc^2$, general relativity, quantum field theory—might already contain what we have been searching for.

9 The Questions, Summarized

1. Vacuum energy exists and is experimentally confirmed. Why was it identified with a geometric constant introduced fifty years earlier, on no stronger basis than mathematical similarity?
2. This identification produced the most severe quantitative discrepancy in the history of physics (10^{120}). After sixty years without resolution, is it worth considering that the identification itself is the error?
3. If we separate Λ (geometry) from vacuum energy (field physics), the cosmological constant problem is reframed and dark energy is re-identified as geometry. What, if anything, do we lose by this separation?
4. If vacuum energy is not Λ , it remains a real, gravitating form of energy. Inside gravitationally bound systems where expansion is absent, its gravity is uncompensated. Is this contribution negligible, or could it be significant?
5. The qualitative properties of gravitating vacuum—invisible, collisionless, cold, stable, ubiquitous, purely gravitational—match every requirement for the missing mass. After forty years of searching for a particle that has not been found, is it possible that what we are looking for has been here all along?

We offer no answers. We believe the questions deserve them.

10 Peer Reviews and Revision History

First Round

Reviewer assessment. The reviewer found the paper clearly written and philosophically interesting, with an admirable scope limitation. Five major concerns were raised: (1) the property table is less constraining than presented; (2) the absolute-vs-relative energy problem is underplayed; (3) the void stability argument is incomplete (Jeans instability); (4) CMB compatibility is a prerequisite, not an optional test; (5) the Bullet Cluster argument is too casual.

Response. We agree with the reviewer on all five points. The paper has been substantially revised to reflect a more modest epistemic position. Rather than attempting to address each concern with calculations or arguments that would inevitably be insufficient for problems this deep, we have restructured the paper as a sequence of questions rather than a sequence of arguments. The property table has been replaced by a qualitative observation of coincidence, explicitly labeled as such. The void stability, CMB, and Bullet Cluster issues are acknowledged as open and serious questions that we do not attempt to answer. The absolute energy problem is noted as genuine and unresolved.

The paper no longer claims to identify the missing mass with vacuum energy. It asks whether there is a reason this identification has not been investigated, and whether the Zel'dovich identification that forestalled it was ever established. We believe these questions stand regardless of whether the answers turn out to favor the proposal.

Second Round

Reviewer assessment. The reviewer found the revision substantially improved and the genre appropriate. Seven minor revisions were requested: (1) state explicitly that the inquiry depends on the unknown vacuum density being of the right magnitude; (2) soften “dissolves” to “is reframed”; (3) restore a distinguishing statement about prior work; (4) add the Bullet Cluster morphology to open challenges; (5) use “estimate” rather than “prediction” for the 10^{120} figure; (6) neutralize the phrasing of Question 3; (7) remove placeholder sections.

Response. All seven revisions have been implemented. (1) A paragraph identifying the unknown vacuum density as the central vulnerability has been added at the end of Section 7. (2) Section 5.1 now reads “is reframed” and includes discussion of the residual question. (3) The prior work paragraph in Section 7 now distinguishes this paper’s approach. (4) The Bullet Cluster morphology is noted as an open challenge in Section 6. (5) “Prediction” has been replaced throughout. (6) Question 3 now reads “What, if anything, do we lose?” (7) Placeholder sections have been removed.

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